

Surprise-Gated Selective Initiation with a Claude-API Arbiter: A Negative Result on Targeted Prompt Revision, with a Mechanism and Safety Diagnostic

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Abstract

Proactive LLM agents need a principled way to decide *when* to act without being asked. We study a surprise-gated selective initiator that routes events selected by a surprise gate to a Claude-API content arbiter, and we ask whether a targeted revision of that arbiter’s prompt improves proactive behavior. At a small curated-trace scope the architecture reaches a hit rate ≥ 0.80 at \$0.0057–0.0118 per true-positive hit, 16–33 $\times$ cheaper per hit than unconditional polling with the same arbiter — a cost reduction that follows mechanically from the gate firing less often. Our main result is negative. Across $N=40$ fresh, externally-authored traces, a pre-registered prompt revision (V_4) designed from three identified residuals is classified *net neutral* relative to the prior prompt (V_2) under the pre-registered decision rule — not a statistical equivalence claim: V_4 -Opus joint-bar failure rate is $11/40 = 27.5\%$ versus V_2 -Opus $10/40 = 25.0\%$ (+2.5 pp, 95% Clopper–Pearson CIs [14.6, 43.9] vs. [12.7, 41.2]). A cell-level mechanism diagnostic explains why: V_4 corrects only 1 of 5 flagged events, because the surprise gate *bypasses the arbiter* on the other four (three auto-surface on negative surprise; one is skipped before the arbiter is consulted). The load-bearing residual is therefore in the loop logic, not the arbiter prompt. We additionally report a safety-relevant side effect: at the strongest tier, V_4 -Opus *over-suppressed* two genuine medical emergencies that the V_2 prompt and the cheaper V_4 tiers surfaced. We are explicit about statistical limits: V_4 and V_2 produce identical joint-bar outcomes on 37/40 traces and identical false-initiation rates on all 40, so the cross-prompt signal rests on three discordant traces (exact two-sided sign test $p = 1.0$); cross-tier quality rankings are reported as cost facts plus exploratory quality signals, not as demonstrated quality advantages. The contributions we stand behind are the mechanism diagnostic, the safety finding, and a pre-registration protocol for LLM-authored evaluation traces with an iterative banned-list extension that structurally controls cross-trace drift.

1 Introduction

A proactive assistant that watches an ambient event stream (inbox, calendar, alerts) must decide, at every moment, whether *this* event is worth interrupting the user for. Two naive policies bracket the design space: *poll* the most capable model on every event (expensive, but never misses a judgement call), or fire on a fixed *timer* (cheap, but blind to content). We study a middle path — a *surprise gate* that forwards only events that deviate from a local predictor’s expectation to a stronger content *arbiter* — and we ask a concrete engineering question:

Given a working surprise-gated initiator, does a targeted revision of the arbiter prompt improve proactive behaviour, and if not, why not?

Our headline answer is *no*, and the *why-not* is *mechanistic*: across $N=40$ externally-authored traces the prompt revision is classified net-neutral under the pre-registered decision rule, and a cell-level diagnostic shows the architecture routes most of the relevant events *around* the arbiter before the prompt can act. This is a more useful result than a marginal win would have been, because it identifies the actual load-bearing surface for future work.

On “surprise” and free energy. We were inspired to test prediction-error gating by the free-energy principle (Friston, 2010), under which prediction error is a principled signal for “something worth attending to.” We want to be precise about what we do *not* claim. The word “surprise” in this paper is an operational engineering construct — the cosine distance between a local predictor’s last prediction and the next observation, embedded by an *independent* embedder (Section 3.2) — and is not a claim about variational free energy, message passing, or expected-free-energy minimisation. Moreover, our polarity sweep (Section 4) *falsifies* the naive high-surprise→act reading: the empirically retained gate surfaces *low*-surprise events. We treat free energy as historical inspiration only; the operative claim is the empirical one.

Contributions.

1. **A negative result with a clean mechanism diagnostic.** A pre-registered prompt revision corrects 1 of 5 flagged events at $N=40$; the other four are bypassed by the surprise gate before the arbiter is consulted. The load-bearing residual is the loop logic, not the prompt (Sections 6, 7).
2. **A safety-relevant finding about prompt revision.** The same revision caused the strongest model tier to over-suppress two genuine medical emergencies that both the baseline prompt and the cheaper tiers surfaced. Targeted NO-clauses traded distractor suppression for high-cost false negatives (Section 8).
3. **A cost characterisation, stated as cost.** At matched arbiter capability the surprise-gated path is 16–33× cheaper per hit than polling, and within a fixed prompt the cheapest Claude tier is ~20× cheaper than the most expensive at no *measured* quality cost — a cost fact, not a demonstrated quality advantage (Section 9).
4. **A pre-registration protocol for LLM-authored evaluation traces** with an iterative banned-list extension that structurally eliminates the cross-trace drift mode that halts a frozen-list protocol (Section 5). We argue this is the most novel methodological piece, distinct from generic pre-registration.
5. **A reproducible artifact bundle** (code, run docs, raw metric JSONs, pricing attestations, per-trace authoring transcripts, analysis summaries) released at the citable tag `arxiv-v1` (Section 12).

2 Related work

Proactive and anticipatory agents. Most LLM-agent frameworks are *reactive*: they act in response to a user turn (Yao et al., 2023) or explore an environment under an externally-supplied objective (Wang et al., 2023). Proactivity — acting without being asked — has a long pre-LLM history in attention-aware computing, where the central problem is the same one we study: deciding when an interruption is worth its cost (Horvitz et al., 1999). Su (2026) revives the idea for LLM

agents via a continuous internal “heartbeat” plus a learned dispatcher, but evaluates a toy LSTM on synthetic categorical traces with no LLM in the loop. We keep the proactive-initiation question and put a real LLM arbiter in the loop, but we deliberately do not benchmark against [Su \(2026\)](#) numerically: the environments and action spaces differ enough that a shared metric would be misleading. We instead build our own deterministic sandbox and report cost-matched comparisons within it (Section 3).

Surprise and novelty detection. Gating on prediction error connects to a large anomaly/novelty-detection literature ([Chandola et al., 2009](#)); embedding-distance novelty is one standard family within it. Our contribution is not a new detector but the finding that, in this proactive-assistant setting, the *useful* polarity is inverted relative to the naive “high-error→attend” intuition, and — more importantly — that the gate’s hard auto-surface/auto-skip bands, not the downstream content classifier, dominate the system’s residual errors.

Empirical rigour in agent evaluation. Agent papers are widely criticised for co-designing the environment, the agent, and the evaluation, making results hard to trust. We adopt pre-registration with locked outcome text and, specifically for the case where evaluation traces are themselves written by LLMs, contribute an iterative banned-list protocol that controls author-side drift (Section 5).

3 Sandbox, agent, and metrics

3.1 Sandbox

We reject stripping prompts from existing embodied benchmarks (it invalidates comparison to their published numbers) and bespoke “living-world” setups (reviewers reasonably suspect tuning). Instead we use a deterministic ambient-assistant sandbox (`sandbox/world.py`, `sandbox/event_trace.py`): a scripted world with an inbox, a calendar, and an injected event stream carrying ground-truth-annotated “should have noticed” events at known simulated times. A *trace* is a list of `Event` objects with a `sim_time`, `content`, `event_id`, an `is_ground_truth` flag, and (for ground-truth events) a keyword tuple and a policy window. Three traces (`dev_v2`, `test_v1`, `test_v2`) were co-developed during architecture search (the *curated* scope); 40 further traces were authored by fresh LLM sessions under the protocol of Section 5 (the *external* scope).

3.2 Architecture (heargent v1)

HEARGENT is a single persistent loop with a fast tick (~ 5 s). Each tick: (1) **observe** new events; (2) **predict** the next observable event with a local qwen2.5:3b-instruct predictor ([Qwen Team, 2024](#)) over a rolling 16-tick window (temperature 0, deterministic); (3) **score surprise** z as the cosine distance between the prior prediction and the new observation, embedded by an *independent* embedder (nomic-embed-text; [Nussbaum et al., 2024](#)) and normalised by a rolling standard deviation; (4) **gate**: a three-way band — $z < z_{\text{surf}}$ *auto-surface*, $z > z_{\text{skip}}$ *auto-skip*, otherwise *consult the arbiter*; (5) **arbiter**: a Claude-API call returning a single-token YES/NO on whether to surface; (6) **log**. Computing surprise with an independent embedder (never the acting model’s own logprobs) is a deliberate guard against the “surprise collapses to the acting model’s perplexity” failure mode.

The hyperparameters z_{surf} , z_{skip} , window = 16, temperature = 0, seed = 42 are frozen from architecture search onward and are *not* retuned in any milestone — a discipline against unfalsifiability-by-

knob-turning. Two frozen agent variants appear: HEARGENTZA ($z_{\text{skip}}=+1.0$) and HEARGENTZA-WIDE ($z_{\text{skip}}=+1.5$); the variant used for each cell is stated explicitly (and is the subject of a baseline caveat in Section 10).

3.3 Scoring metric and its statistical treatment

A trace cell *fails* the **joint bar** iff

$$\text{hit_rate} < 0.80 \quad \text{OR} \quad \text{false_initiation_rate_per_hour} > 5.0/\text{h}.$$

We pre-registered this single binary predicate as the primary outcome and use it uniformly. We are explicit about two consequences a careful reader will raise. *First*, the predicate collapses two continuous quantities to one bit per trace; we treat the traces as independent Bernoulli trials (they are fresh-session-authored under a cumulative banned list) for Clopper–Pearson (Clopper and Pearson, 1934) and bootstrap (Efron and Tibshirani, 1993) interval estimation. This is conservative — it discards within-trace gradation — but it also *costs power*, which is why we add continuous and paired secondary analyses in Section 6 rather than relying on the binary count alone. *Second*, the thresholds 0.80 and 5.0/h are design choices; we report a sensitivity analysis over them (Table 2).

Cost metric. We report *dollar-cost-per-hit* (\$/hit): Claude-API arbiter spend divided by true-positive initiations in the cell. Zero-hit cells are excluded from per-cell aggregates. Cross-method cost comparisons use a *matched-arbiter* denominator (gated-tier vs. poll-tier at the same Claude model), so the comparison is held fixed on the model dimension.

4 Curated-scope result and ablations

Before the external evaluation, we establish that both the surprise signal and the content arbiter are load-bearing on the curated traces, and that the cost advantage is real.

The surprise signal matters, with inverted polarity. A random gate matched to HEARGENT’s firing rate equals or beats the naive forward gate (surface high-surprise) on hit/false. The *inverted* gate (surface *low*-surprise, $z_{\text{surf}}=-0.5$) beats matched random on both reported axes (hit and false/h) on `test_v1` and transfers under a frozen threshold. This is the empirical falsification of the naive free-energy reading noted in Section 1: the operative signal is “less surprising than the local predictor expected,” which we interpret as the predictor being confidently wrong in a way that flags a systematic gap. We adopt the inverted polarity empirically and do not claim it as evidence for free energy.

The content signal matters. A 20-seed comparison against a matched-firing-rate random arbiter has the content arbiter beating random on 18/20 seeds on `test_v2` (meeting a pre-registered 90% bar) and on false/h in 20/20 seeds on `test_v1` ($\Delta\text{false/h} = -8.08 \pm 2.98$).

Cost advantage (stated as cost). On the curated traces the surprise-gated path reaches $\text{hit} \geq 0.80$ at \$0.0057–0.0118 per hit, 16–33× cheaper than polling the same arbiter on every event. We emphasise that this ratio follows *mechanically* from the gate forwarding fewer events; the non-trivial content of the result is that hit rate is *maintained* at the lower call volume, within the sampling limits we make explicit below.

The residual that motivated everything downstream. The single remaining curated-scope miss is `package_arrival` on `test_v1`: a bare delivery notification outside the arbiter prompt’s closed YES-enumeration. This foreshadows the external-scope finding: the prompt’s YES list is *closed*, so any ground-truth event outside it is a coverage gap rather than a tunable threshold. Section 5 stresses exactly this coverage with externally-authored traces.

5 External-authoring protocol

Why. Curated traces close several reviewer attacks (random-gate ablation, seed variance, polarity transfer) but not the “you designed your own eval” attack. We therefore have evaluation traces written by *fresh* LLM sessions with no access to the agent code, the arbiter prompt, the surprise scorer, or any prior run document, under a pre-registered spec.

Four-commit shape. Every external milestone follows: **Commit A** — pre-registration with frozen design choices, locked bars, and a pre-drafted outcome paragraph per possible result with integer placeholders only; **Commit B** — code changes (if any) plus bit-identical carry-forward re-runs against prior JSONs as pass gates; **Commit C** — fresh-session trace authoring behind an audit gate (structural checks, banned-list checks, ≥ 8 -word verbatim-overlap checks against all prior traces); **Commit D** — evaluation, mechanical outcome-row identification under first-match-wins precedence, and verbatim fill-in of the locked paragraph. Wording is never edited after Commit A. This precludes post-hoc reframing of a milestone’s outcome; the locked paragraphs are reproduced in Appendix C.

The novel piece: iterative banned-list extension. A *frozen* banned-list protocol drifts: across six fresh sessions our frozen-list milestone accumulated 9+ thematic and 2 literal-ID cross-trace overlaps with worsening severity, and halted at $N=3$. The halt is itself the finding — the frozen list cannot scale. We then revise the protocol: after each accepted trace, its event-ids, themes, and keyword tuples are appended to the banned list, and the next fresh session sees the extended list. Under pre-registered mechanical success criteria (zero literal-ID collisions; zero keyword-tuple byte-wise matches; zero ≥ 8 -word verbatim phrase matches in accepted traces), the iteratively-extended protocol reaches its targets with the banned list growing monotonically to 217 ids / 122 themes / 121 tuples by the last external milestone. This structural control of author-side drift — not pre-registration per se, which is standard hygiene — is what we offer as a methodological contribution for LLM-authored evaluation.

Cross-protocol caveat. The combined $N=40$ sample spans several protocol generations (frozen and iterative). Findings stable across generations are more robust than within-generation ones; the cumulative banned list ensures within-protocol prior-trace contamination is structurally precluded. A residual validity limit — that all authors are Claude-family models — is stated in Section 10.

6 The negative result: V4 prompt revision is classified net neutral at $N=40$

Setup. The baseline arbiter prompt (*V2*) is a closed enumeration of YES/NO classes. From three identified residuals — a YES-bias on three distractor classes (retail back-in-stock, recurring calendar suggestions, casual social meetups) shared across model tiers, plus a coverage gap on

discretionary-deadline obligations — we pre-registered a single revision (V_4) that adds three explicit NO subclauses and one extended YES class. The full V_4 text (1851 bytes; sha256 09be309d...) is frozen and reproduced in Appendix A. V_4 was run single-shot with no iteration reserve, against a mechanically-determined outcome row.

Primary outcome. Table 1 reports per-tier joint-bar failure rates at combined $N=40$. The pre-registered comparison is V_4 -Opus vs. V_2 -Opus.

Cell	N	fails	rate	95% CP CI	CP width	\$/hit
V4-Opus	40	11	27.5%	[14.6%, 43.9%]	29.3 pp	\$0.1475
V4-Sonnet	40	9	22.5%	[10.8%, 38.5%]	27.6 pp	\$0.0194
V4-Haiku	40	9	22.5%	[10.8%, 38.5%]	27.6 pp	\$0.0065
V2-Opus	40	10	25.0%	[12.7%, 41.2%]	28.5 pp	\$0.1186

Table 1: Per-tier joint-bar failure rates at combined $N=40$ on $\{\text{test_v4}..v8, \text{test_v11}..v15, \text{test_v21}..v50\}$. Bootstrap CIs (seed 42, 2000 resamples) agree with Clopper–Pearson within 5 pp at both bounds. Source: `runs/data/22d-analysis-summary.json`.

The pre-registered predicate composition resolves mechanically: the V_4 -vs- V_2 point-estimate delta is +2.5 pp (NEUTRAL, $|\cdot| \leq 10$ pp); the V_4 -Opus CI width 29.3 pp is ≤ 30 pp ($\neg$ WIDE_CI); mechanism is PARTIAL (1/5, Section 7); and the no-regression bar is broken (V_4 -Opus compliant fails 10 vs. V_2 -Opus 8). This fires the pre-registered row **“mechanism-only, net neutral.”** We stress that this is a *classification under the pre-registered decision rule*, not a statistical equivalence claim: the data cannot establish that V_4 and V_2 are equivalent, only that the pre-registered comparison detected no improvement (and a point estimate mildly favouring V_2). V_4 ’s mechanism intervention is partly real but is offset by a compensating regression on compliant content, leaving the net failure rate unchanged (in fact +2.5 pp on the wrong side of zero at the point estimate).

Secondary analyses (not pre-registered, paired/continuous). Because the binary joint bar discards information and inflates apparent under-power, we add paired trace-level analyses (all computed from `runs/data/22d-analysis-summary.json`):

- **V4 and V2 are identical on most traces.** V_4 -Opus and V_2 -Opus produce the *same* joint-bar outcome on 37 of 40 traces, and *identical* false-initiation rates on all 40. The entire cross-prompt difference lives on the hit side, on three discordant traces: V_4 better on `test_v8`; V_2 better on `test_v38` and `test_v50`. An exact two-sided sign test on these 3 discordant pairs gives $p = 1.0$.
- **Continuous hit-rate difference.** Mean $\Delta_{\text{hit}} (V_4 - V_2) = -0.02$, median 0.0; non-zero on 7 of 40 traces, ranging -0.40 to $+0.20$. **Continuous false-rate difference:** exactly 0.0 on every trace.
- **The under-power is structural, not just small- N .** A two-proportion test at $N=40$ per arm and base rate $p \approx 0.25$ has 80% power only for differences of roughly 25–30 pp; resolving a ± 10 pp difference would need several hundred traces per arm. More to the point, with the two prompts agreeing on 37/40 traces, the signal rests on three observations and no test on it can be conclusive.

Threshold sensitivity. Table 2 recomputes the fail counts under varied joint-bar thresholds. The *absolute* rate is threshold-sensitive (e.g. tightening the hit bar to 0.85 roughly doubles failures),

which is exactly why we caution against over-reading the headline percentage. But the *V4-vs-V2 relationship* is stable: V4-Opus fails $\geq$ V2-Opus by 0–4 traces at every setting, never reversing. The qualitative conclusion (V4 net-neutral-to-slightly-worse) is robust to the threshold choice; the absolute level is not, and we do not claim it is.

Threshold (hit < / false/h >)	V4-Opus fails	V2-Opus fails
0.75 / 4.0	20	20
0.75 / 5.0	11	10
0.75 / 6.0	11	10
0.80 / 4.0	20	20
0.80 / 5.0 (<i>pre-registered</i>)	11	10
0.80 / 6.0	11	10
0.85 / 4.0	29	25
0.85 / 5.0	24	20
0.85 / 6.0	24	20

Table 2: Threshold sensitivity of the joint-bar fail counts (out of 40). The absolute level moves a lot; the $V4 \geq V2$ ordering does not.

What the negative result says. A pre-registered, mechanism-motivated prompt revision did not net-improve the system under the pre-registered $N=40$ decision rule, and the direction of the small residual is unfavourable. This is a clean falsification of the hypothesis that targeted prompt revision is the next useful intervention for this workload. The next two sections explain the mechanism and surface a safety-relevant cost of the revision.

7 Mechanism diagnostic: the surprise gate bypasses the arbiter

The negative result is not a black box. Table 3 traces each of the five pre-registered flagged events through the loop for V2-Opus and V4-Opus.

Event	Trace	V2-Opus path	V4-Opus path	V4 fix?
trivia_league_round	test_v11	$z < -1$ auto-surface	$z < -1$ auto-surface	—
grocer_back_in_stock	test_v12	$z < -1$ auto-surface	$z < -1$ auto-surface	—
calendar_yoga_suggest	test_v12	$z < -1$ auto-surface	$z < -1$ auto-surface	—
mom_birthday_heads_up	test_v8	surprise-gate skip	surprise-gate skip	—
bridgers_presale_window	test_v8	arbiter NO (wrong)	arbiter YES	✓

Table 3: Cell-level mechanism check on the five flagged events. V4 corrects exactly one — the only one where the arbiter is actually consulted.

The four unfixed events share one mechanism. V4’s prompt clauses can only act when the arbiter is consulted. On three flagged events the surprise gate *auto-surfaces* the event on strong negative surprise ($z < -1$) before any arbiter call, so V4’s NO subclauses — which were *intended* to suppress exactly these distractor classes — are not load-bearing: the prompt never sees the event. (We flag a wording bug in the prior draft here: those NO subclauses do not “correctly suppress” the distractors; they are bypassed by the gate, which is the whole point of this diagnostic.) On the fourth event the gate *skips* the event entirely (the local predictor expected it), so again the

arbiter is never consulted and V4’s extended-YES clause has no opportunity to fire. Only the fifth event (`bridgers_presale_window`) is arbiter-dispatched, and there V4’s extended YES enumeration correctly flips V2’s erroneous NO.

Consequence for future work. The load-bearing residual at $N=40$ is the *surprise-gate-bypass family* — a loop-logic surface in `agent/loop.py`, not the arbiter prompt. A further prompt revision (V5) could affect at most the single arbiter-dispatched cell. The remaining four require a loop-logic intervention: route the $z < -1$ auto-surface band through the arbiter, narrow the auto-surface band, or add hierarchical routing that sends a wider event band to the arbiter. We pre-registered this residual class as a non-goal of the present milestone and named surprise-gate retuning as its own future milestone before seeing these results (Appendix C), so this is a pre-registered diagnostic, not a post-hoc reframe.

8 Safety finding: V4-Opus over-suppresses high-priority content

The most consequential side effect of the V4 revision is not in the headline rate; it is in *which* traces moved. The two traces where V4-Opus is worse than V2-Opus are both high-stakes:

- `test_v38`: an elderly parent in septic shock (urosepsis, ICU admission) — a genuine medical emergency. V2-Opus surfaced it; V4-Opus did not.
- `test_v50`: a household member with an acute large-vessel-occlusion ischemic stroke (tPA-eligibility / thrombectomy workup), plus an ESPP enrollment-window financial deadline. V4-Opus over-suppressed; V2-Opus surfaced.

Crucially, this is *tier-stratified*: V4-Sonnet and V4-Haiku surfaced these items; only V4-Opus, the strongest tier, over-suppressed them. Our reading is that the V4 prompt *as a whole* — longer and more exclusion-oriented than V2 — shifts the strongest tier toward suppression, so that Opus applies the added exclusion reasoning even to genuinely-YES content. We are careful here: the present data do not isolate *which* added clause is responsible (the extended-YES clause is in fact the one *validated* addition, Section 7), only that the V4 prompt collectively produces the over-suppression at the Opus tier. We present this as the paper’s central cautionary finding: in a proactive assistant, adding exclusion clauses to a prompt can trade cheap false positives for expensive false negatives, and the trade can be worse at a stronger model tier. We state plainly that this rests on two traces and is therefore an existence proof with a hypothesised mechanism, not a quantified rate; but an existence proof of a safety regression is exactly the kind of finding a single instance suffices to establish.

9 Cost and cross-tier analysis

We separate two things that the prior draft conflated: *cost*, which our data establish firmly, and *cross-tier quality*, which it does not.

Cost facts (firm). Cost-per-hit is deterministic from token pricing. At matched arbiter capability the gated path is 16–33× cheaper per hit than polling across milestones. Within the V4 prompt at $N=40$, V4-Haiku (\$0.0065/hit) is 22.8× cheaper than V4-Opus (\$0.1475/hit) and V4-Sonnet (\$0.0194/hit) is 7.6× cheaper; V4 is 1.24× more expensive per hit than V2 within tier, reflecting the longer prompt. These ratios are not subject to sampling error in the way failure rates are.

Cross-tier quality (exploratory, not demonstrated). At $N=40$, V4-Haiku and V4-Sonnet both show a point-estimate failure rate 5 pp *below* V4-Opus (22.5% vs. 27.5%), driven entirely by the two over-suppression traces of Section 8 where the cheaper tiers surfaced and Opus did not. We deliberately do *not* call this “Pareto-dominance.” The quality gap is two discordant traces; the tiers’ CIs overlap heavily; an exact two-sided sign test on the Opus-vs-Haiku discordance (2 pairs, both favouring Haiku) gives $p = 0.5$. The defensible statement is: *there is no measured quality justification for the Opus cost premium under V4 in this sample*, and the cheaper tiers are preferable on **cost** grounds. We treat the tier quality ranking as an exploratory signal motivating the cross-tier routing milestone named in Section 11, not as an established result.

Confirmatory vs. exploratory. To be unambiguous: the single confirmatory outcome is the pre-registered V4-vs-V2 row (Section 6) plus its pre-registered mechanism and cross-tier branches. All tier rankings, deployment preferences, per-event safety interpretations, and Pareto statements are exploratory and are not corrected for multiple comparisons. We report them for transparency and to direct future work, not as confirmed findings.

Pareto figure. Figure 1 plots cost-per-hit (log scale) against joint-bar failure rate at $N=40$ for the four cells with $N=40$ coverage, with 95% Clopper–Pearson error bars. The cheaper tiers sit far to the left at a comparable-or-lower point-estimate failure rate, but the error bars make the central caveat visible: the V4-Opus and V2-Opus intervals overlap almost entirely, so the *quality* axis is not discriminated at this N — the firm result is the horizontal (cost) gap, not the vertical (quality) one.

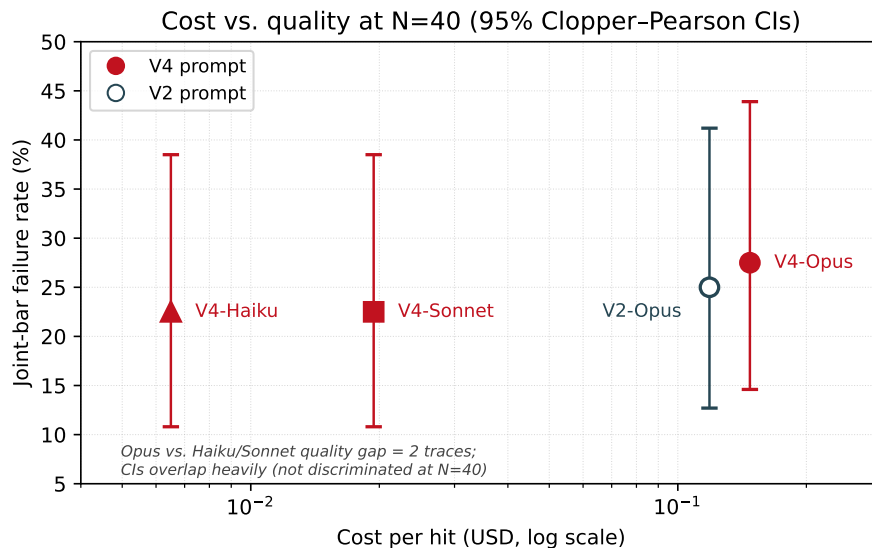


Figure 1: Cost vs. quality at combined $N=40$. Filled markers: V4 prompt; open markers: V2 prompt. x -axis is dollar-cost-per-hit (log scale); y -axis is joint-bar failure rate with 95% Clopper–Pearson intervals. The cheaper Claude tiers (Sonnet, Haiku) are an order of magnitude cheaper per hit than Opus; the failure-rate differences among all four cells are within sampling error (overlapping CIs). Data: runs/data/22d-analysis-summary.json.

10 Limitations

Statistical power. The cross-prompt comparison is under-powered in a specific, structural sense: V4 and V2 agree on 37/40 joint-bar outcomes and on all 40 false rates, so the signal rests on three discordant traces. No re-analysis of these data can make a three-observation difference conclusive; only a substantially larger or more discriminating evaluation can. We therefore present the cross-prompt result as net-neutral-with-mechanism, not as a measured V4-vs-V2 effect size.

Baseline variant mix (resolved by sensitivity run). The $N=40$ V2-Opus baseline combined two agent variants — 10 cells (`test_v4..v15`) at HEARGENTZAWIDE ($z_{\text{skip}}+1.5$) and 30 cells (`test_v21..v50`) at HEARGENTZA ($z_{\text{skip}}+1.0$) — while all V4 cells use HEARGENTZA uniformly. We ran the corresponding sensitivity check, re-deriving the 10 `test_v4..v15` cells under V2-Opus at uniform $z_{\text{skip}}+1.0$ (`runs/data/restate-v2opus-za-*.json`). **The uniform V2-Opus failure rate is identically** $10/40 = 25.0\%$: no cell changes joint-bar state, and Δ_{PE} vs. V4-Opus is unchanged at +2.5 pp. The re-run reproduced the prior per-cell hit/false values on 9 of the 10 cells; the one difference (`test_v11` hit $1.00 \rightarrow 0.80$, still passing the bar) is the expected effect of the narrower skip band and confirms the premise that $z_{\text{skip}}+1.5$ yields slightly higher hit rates. The Row 2 net-neutral verdict is thus *empirically* robust to the baseline variant choice, not merely argued to be. (This is consistent with the structural observation that Row 2 fires on $\neg \text{WIDE_CI}$, a function of the V4-Opus CI width, independent of the V2 baseline variant.)

Same-model-family authoring. The externally-authored traces are external to the agent code and arbiter prompt but *not* external to the Claude model family: fresh-session separation and the banned-list extension control literal contamination, not possible distributional alignment between a Claude author and a Claude arbiter. Human-authored and cross-vendor-authored traces are required before claiming broader external validity, and are named future work.

Single sandbox, single vendor. All measurements are on one ambient-assistant sandbox and one vendor’s tier ladder; no cross-domain or cross-vendor (e.g. GPT-class, Gemini-class, Llama-class) comparison is included. The protocol stabilises *within-protocol* trace variance; it does not address cross-domain variance.

Determinism is epoch-bounded. Claude-API arbiter calls are at temperature 0 and are decision-equivalent *within an Anthropic-side state epoch*. We observed at least one cross-epoch drift event (a trace’s hit rate moving $1.0 \rightarrow 0.8$ over a five-day window). We therefore claim reproducibility of the integer joint-bar counts, not of every per-event decision across arbitrary dates. Our drift smoke (Appendix B) is how we detect such shifts.

Single-shot pre-registration. The no-iteration-reserve posture closes the iteration-budget attack but means any revision identified after a milestone closes (e.g. a V5) is a separate milestone, not a within-paper iteration. This is a defensibility feature and a throughput constraint.

11 Future work

Each item is named here so reviewers can confirm it is not pulled into this paper; each has its own pre-registration scope.

1. **Surprise-gate retuning** — the load-bearing residual (Section 7). Route or narrow the $z < -1$ auto-surface band and the skip path in `agent/loop.py` so the arbiter is consulted on the bypassed events. Addresses 4 of the 5 flagged events.
2. **Cross-tier routing** — local→Sonnet/Opus escalation; the cost gap with no *measured* quality penalty makes a cheaper-tier default attractive, pending the quality question being settled at higher power.
3. **Cross-vendor and human-authored traces** — to address the same-model-family and single-vendor limits.
4. **V5 prompt revision** — conditional on (1) closing the bypass residual first. At $N=40$ the V4 prompt is two-sided: its extended-YES clause produced the one validated correction (the single arbiter-dispatched cell; Section 7), while the prompt *as a whole* drives the Opus over-suppression of high-priority content (Section 8). Since the present data do not isolate the responsible clause, a V5 should be guided by a per-clause ablation that keeps the validated correction and removes the over-suppression — the pre-registered remediation hypothesis (revert the YES extension, Appendix C) is one candidate to test, not an established cause. V5 is in any case the wrong next step until the surprise-gate bypass is addressed.

12 Reproducibility and artifact release

Bundle. The artifact ships at the citable tag `arxiv-v1` (immutable commit 0005499) on <https://github.com/padman667/heargent>, license MIT (LICENSE at the repo root). It contains: the code-base (`sandbox/`, `agent/`, `baselines/`, `eval/`); per-milestone run docs (`runs/01-runs/22`); the raw metric JSONs (`runs/data/`); per-milestone pricing attestations (`*-pricing-attestation-*.json`); per-trace authoring transcripts (`22c-author-test_v*.json` and earlier equivalents); and the mechanically-computed analysis summary for the binding $N=40$ result, `22d-analysis-summary.json`.

Reproduction. Short form for one $N=40$ cell:

```
git clone https://github.com/padman667/heargent
cd heargent && git checkout arxiv-v1
uv sync
ollama pull qwen2.5:3b-instruct
ollama pull nomic-embed-text
# set ANTHROPIC_API_KEY, then:
uv run python -m eval.run_trace \
  --agent agent.loop:HeargentZA \
  --arbiter-mode claude \
  --arbiter-system-prompt v4 \
  --arbiter-model haiku \
  --trace test_v31 \
  --out runs/data/22d-haiku-v4-test_v31.repro.json
# ... 60 V4 cells (3 tiers x 20 traces) + 20 V2-Opus cells
```

Cost to reproduce. The full $N=40$ harness — the 60-cell V4 sweep (Opus, Sonnet, Haiku $\times$ 20 traces) plus the 20-cell V2-Opus extension — costs \$8.6 of Claude-API spend at the paper’s locked internal modeling rate (Opus \$15/\$75 per MTok, retained for cross-milestone consistency; `runs/data/22a-pricing-attestation-2026-05-22.json`), or roughly \$3–4 at the published rate as

of 2026-05 (Opus \$5/\$25 per MTok). The Haiku-only V4 path reproduces the cheapest tier for ~\$0.20. The local predictor and embedder are free.

13 Conclusion

A surprise-gated proactive agent with a Claude-API arbiter is cheap, and a careful, pre-registered revision of the arbiter prompt did not net-improve it. The reason is mechanistic and, we think, the real contribution: the surprise gate routes most of the relevant events around the arbiter, so the prompt is not the load-bearing surface. Along the way the revision introduced a safety-relevant regression at the strongest tier — over-suppressing genuine medical emergencies — which is a cautionary result about prompt-level edits in proactive systems. We have been explicit about where the statistics are thin (three discordant traces carry the cross-prompt signal) and have framed cross-tier comparisons as cost facts rather than quality claims. The pre-registration protocol with iterative banned-list extension is, we argue, a reusable contribution for any evaluation whose traces are written by LLMs.

Acknowledgments

Anthropic’s Claude models were used as the arbiter, as fresh-session trace authors, and as a code-editing tool throughout. Per-session and per-trace attribution is recorded in the milestone run docs.

A V4 arbiter prompt (verbatim)

SHA-256 of the prompt text: 09be309de5609a3c599d401e93ee4b35e655be24232ece6570a51893f80f56a6 (1851 bytes). V2’s six YES classes and four NO classes are retained verbatim; three explicit NO subclauses and one “discretionary-deadline obligation” YES class are added.

```
ARBITER_SYSTEM_PROMPT_V4 = (  
    "You are a triage filter for a proactive assistant. Decide whether a\n"  
    "single observed event warrants surfacing to the user right now.\n"  
    "\n"  
    "Surface (YES) if the event describes any of:\n"  
    " - an urgent safety or security issue (fire, break-in, medical\n"  
    " emergency, security breach, unauthorized access)\n"  
    " - a schedule change affecting the user personally (meeting moved,\n"  
    " meeting cancelled, flight delayed, appointment rescheduled,\n"  
    " appointment cancelled)\n"  
    " - a financial or deadline obligation the user must act on within\n"  
    " the next few days (bill due, rent due, report deadline,\n"  
    " payment reminder)\n"  
    " - a message or delivery directed personally to the user (package\n"  
    " delivered, callback requested, voicemail or phone message,\n"  
    " hospital calling about a family member)\n"  
    " - a weather alert or external condition that would plausibly\n"  
    " change the user’s planned day\n"  
    " - a production/on-call alert or outage for a system the user owns\n"  
    " - a discretionary-deadline obligation: a family milestone with\n"  
    " social cost (birthday or anniversary heads-up) OR a\n"  
    " scarcity-bounded opportunity with a limited window (presale or\n"  
    " queue offer with a ticket limit and a window under one hour)\n"  
    "\n"
```

```

"Do NOT surface (NO) if the event is:\n"
" - routine status, uptime, heartbeat, or \"all systems normal\" pings\n"
" - marketing, promotional, or newsletter content\n"
" - generic daily briefings that explicitly state no urgent items\n"
" - feature announcements, app updates, or social/channel invites\n"
" - a back-in-stock or restock notification without an explicit\n"
" scarcity window (e.g., \"your favorite item is back in stock\")\n"
" - a recurring-event calendar suggestion or app suggestion\n"
" (e.g., \"add as recurring?\")\n"
" - a casual social-meetup notification without time-pressure\n"
" (e.g., \"pub trivia tonight at 7pm\")\n"
"\n"
"Output exactly YES or NO, uppercase, on a single line. No explanation."
)

```

B Milestone chronology (audit map)

The main text is organised by finding. For audit, the empirical work proceeded through the following milestones; each terminates in a locked outcome paragraph (Appendix C) and is reproducible from the run doc of the same name.

Milestone	Run doc	Scope / lever	One-line outcome
M2–M6a	runs/05–12	architecture search	inverted gate + banded arbiter frozen
M7	runs/13	20-seed random-arbiter	content signal load-bearing
M8b	runs/15	first external trace	V2 coverage gap surfaced (hit 0.40)
M9–M10	runs/16–17	model-scale (Opus)	scale closes some coverage gaps
M10b	runs/18	$N=5$ frozen list	halts at $N=3$ (frozen-list drift)
M11a	runs/19	$N=5$ iterative list	drift controlled; partial closure
M11b	runs/20	cross-tier sweep	YES-bias is prompt-inherent, not scale
M11a-ext	runs/21	V4 prompt, $N=20$	underpowered; V4 partial 1/5
M11a-ext-ext	runs/22	V4 prompt, $N=40$	net neutral; binding result

Table 4: Milestone audit map. The binding empirical result is the $N=40$ row (runs/22); the $N=20$ row is the intermediate under-powered checkpoint it supersedes. A triangulated bit-identical drift smoke runs at each Commit D.

C Locked outcome paragraphs (verbatim, pre-registered)

For audit, the pre-registered outcome paragraph filled at the binding $N=40$ Commit D is reproduced verbatim below; the surprise-gate-bypass residual was pre-registered as a non-goal with a named follow-up milestone *before* the data were written.

Across $N=40$ fresh externally-authored traces, V4-Opus failure rate $11/40 = 27.5\%$ (95% CP CI [14.6%, 43.9%]; width 29.3 pp) is within ± 10 pp of V2-Opus $10/40 = 25.0\%$ (delta +2.5 pp; $\neg$ wide-CI at $N=40$). V4 mechanism check passes 1 of 5. V4-Opus compliant-content failure count = 10 vs. V2-Opus 8 (+2 traces) breaks the no-regression bar. V4 prompt revision fixes

the targeted mechanism cell but introduces compensating regression on compliant content; net failure rate unchanged. V5 should retain V4’s NO additions while reverting the YES extension. The surprise-gate-bypass residual remains the load-bearing mode; surprise-gate retuning, not a further prompt revision, is the appropriate next intervention.

The remaining pre-registered outcome paragraphs for all milestones are reproduced verbatim in the artifact run docs (`runs/14–runs/22`).

D Reviewer-vulnerable surfaces

Condensed from the per-milestone pre-registrations. (1) *p-hacking on the trace distribution*: the iteratively-extended banned list and its mechanical drift criteria are published with the artifact. (2) *author–arbiter contamination*: fresh sessions at a separate working directory, audited zero-context output; deeper same-family alignment is owned as a limitation (Section 10). (3) *multiple comparisons*: one confirmatory outcome row; all rankings exploratory (Section 9). (4) *cost interpretation under unequal hit rates*: matched-arbiter denominator, zero-hit cells excluded. (5) *determinism*: epoch-bounded decision-equivalence with a drift smoke, not universal byte-determinism (Section 10).

Risk register. The three pre-registered project risks and their mitigations: (R1) surprise-collapses-to-perplexity — independent embedder, not the acting model’s logprobs; (R2) a strong baseline matches hit rate — embraced as the framing (selective initiation at lower cost); (R3) unfalsifiability-by-knob-turning — frozen hyperparameters, locked outcome paragraphs, single-shot posture.

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